TESLARU ROLAND

TEMA 7

Teoria Sistemelor

1. Considerând acelaşi sistem de la exemplul 1, $G_d(s) = \dfrac{10}{s^2 + 14s + 50}$, să se construiască schema Simulink din figura 7.6. şi să se vizualizeze semnalele obţinute.

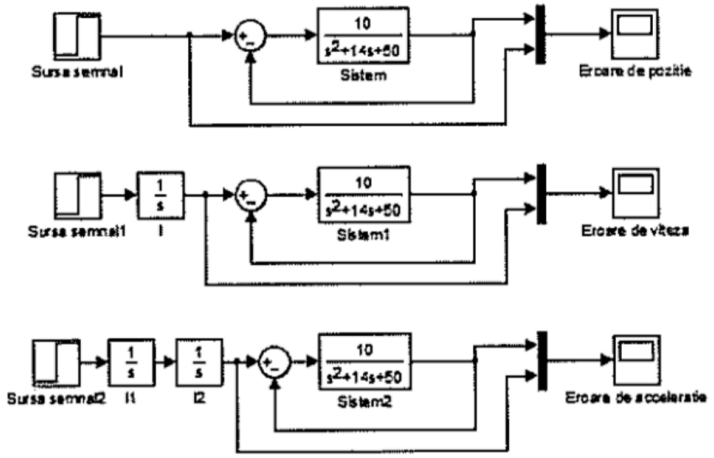
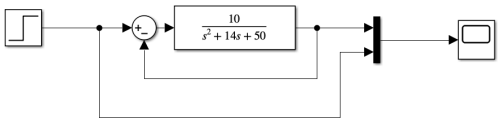
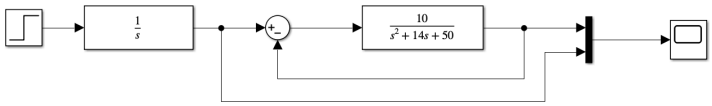
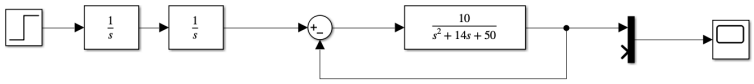


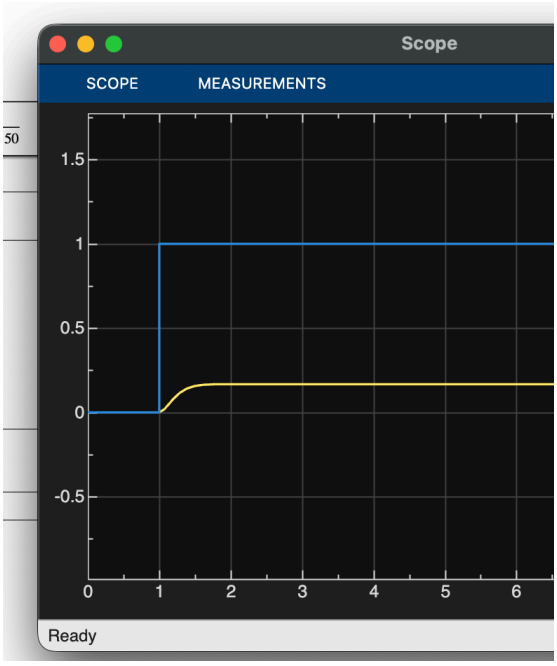
Fig. 7.6 Tipurile de erori staţionare



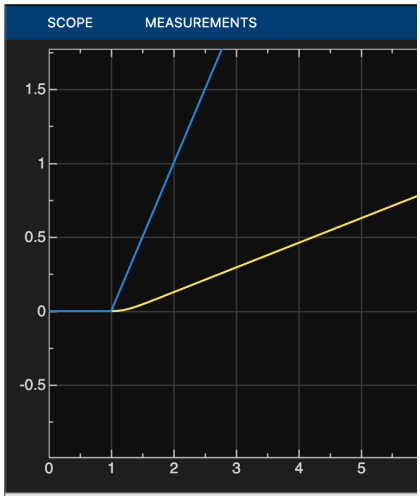




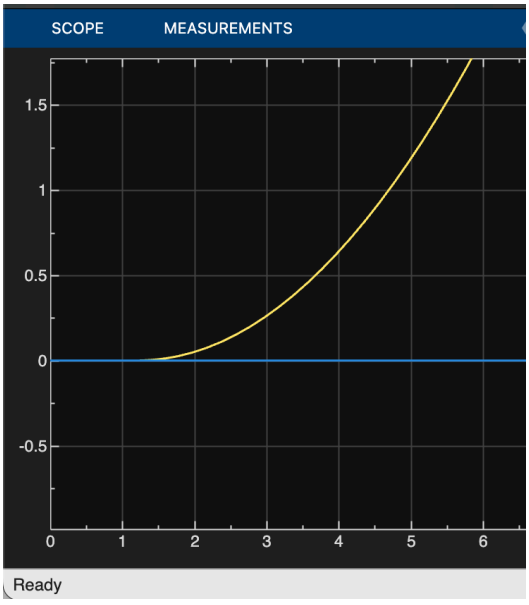
Primul Scope:



Al doilea Scope:



Al treilea Scope:



3. Să se simuleze în Simulink elementele de întârziere de ordin I şi ordin II, pentru semnale de intrare de tip sinusoidal având pulsaţiile $\omega = 1,5,10,20,100$ rad/sec şi amplitudinea $A_i=10$. Să se afişeze pe acelaşi osciloscop semnalele de la ieşirea sistemului pentru valori ale lui $\zeta = 0.1,0.4,0.707,1,2$. Să se noteze valorile amplitudinii A_e pentru fiecare pulsaţie ω şi fiecare factor de amortizare ζ. În final, să se traseze grafic funcţia $f(\omega) = \frac{A_e}{A_i}$.

$$G_d(s) = \frac{s+2}{s^2+5s+9}$$

$s^2+5s+9 \rightarrow$ nu putem factoriza cu un s

$\Rightarrow g=0 \rightarrow$ de tip 0

$$\varepsilon_{stp} = \frac{1}{1+K_p} \qquad K_p = \lim_{s\to 0} G_d(s)$$

$$K_p = \lim_{s\to 0} \frac{s+2}{s^2+5s+9} = \frac{2}{9}$$

$$\varepsilon_{stp} = \frac{1}{9/1+2/9} = \frac{1}{11/9} = 1 \cdot 9/11 = 9/11 \Rightarrow \varepsilon_{stp} = 9/11$$